

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Concept Mapping Questions

Course Code:	CSA0315	Course Title:	Data Structures
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Mapping Questions
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays&Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
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Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 3 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C'	Concept mapping of Basics in Data Structure	1	30
Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.	Fibonacci series	2	40
Primitive and non-primitive data structure, Linear and non-linear data structure Arrays	Concept mapping on classifications of data structures.	3	40
GRAND TOTAL:			100 Marks

Annexure A – Application Based Assessment

- 1) Write a c program and draw the flow chart as concept map for finding the count of odd and even numbers in an array of n numbers.
- 2) Write a concept map based flow chart and also write the program to generate the Fibonacci series and sum of the series up to n numbers.
- 3) A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

11.07.26

V. Nandha Krishna

Assignment

19.2524012

CSA0315 Data Structure
Dr. T. Rajith Kumar

- 1) Write a C program to count the number of odd and even numbers in an array of n

```
#include <stdio.h>
```

```
int main ()
```

```
{
    int n, i, arr[100], even=0, odd=0;
```

```
    printf("Enter the number of elements: ");
    scanf("%d", &n);
```

```
    printf("Enter %d elements: \n", n);
    for (i=0; i<n; i++)
```

```
{
    scanf("%d", &arr[i]);
    if (arr[i] % 2 == 0)
        even++;
```

```
    else
```

```
        odd++;
```

```
}
```

```
printf("Number of Even numbers = %d \n", even);
```

```
printf("Number of odd elements = %d \n", odd);
```

```
return 0;
```

```
}
```

read n (size of array)

Initialize i=0
odd=0, even=0

read arr[i]

arr[i] % 2 == 0

Yes

even++

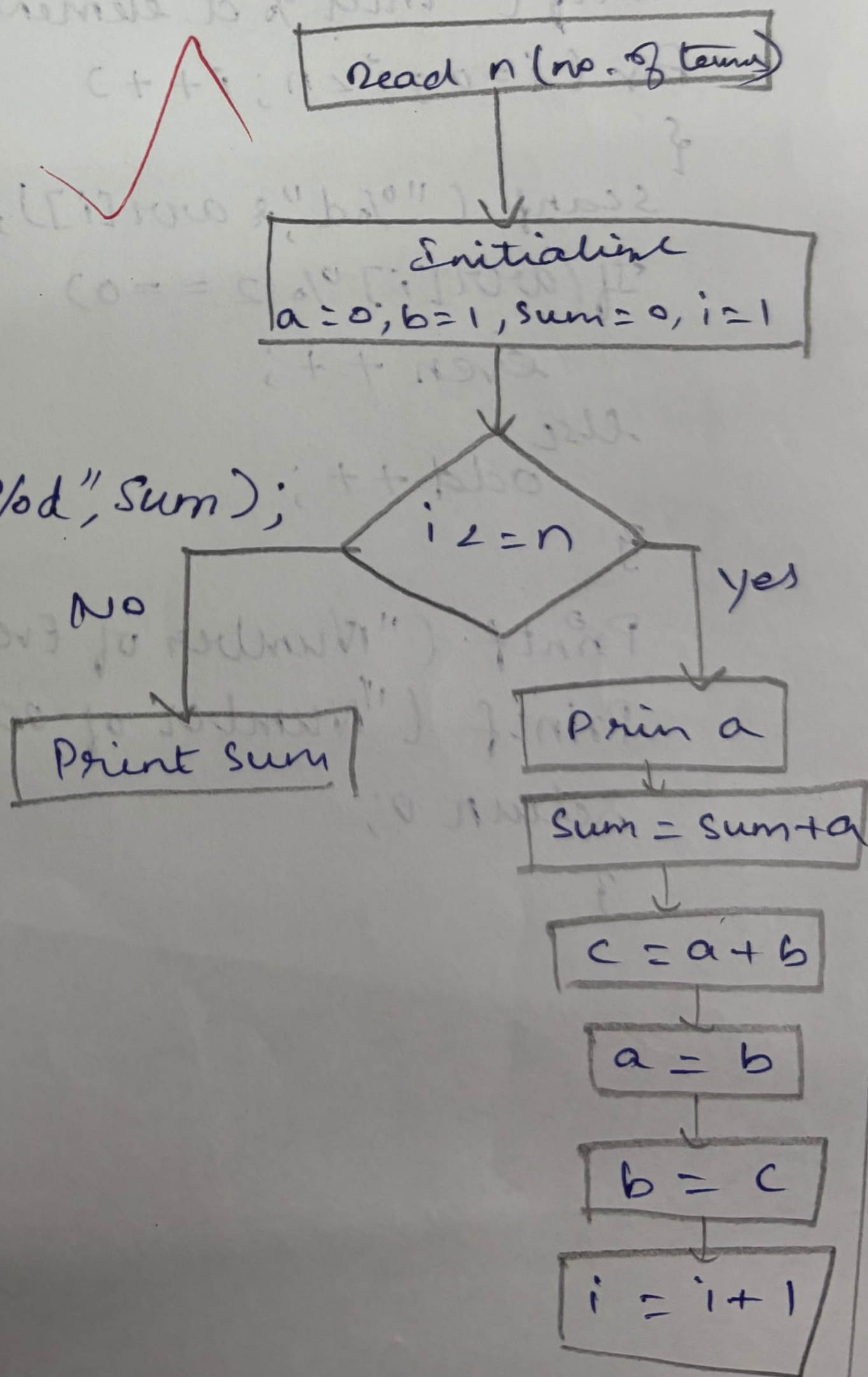
No

odd++

Print count

2) Write a C program to print Fibonacci series and its sum

```
#include <stdio.h>
int main ()
{
    int n, i, a=0, b=1, c, Sum=0;
    printf ("Enter the number of terms: ");
    scanf ("%d", &n);
    printf ("Fibonacci series: ");
    for (i=1; i<=n; i++)
    {
        printf ("%d", a);
        Sum += a;
        c = a+b;
        a = b;
        b = c;
    }
    printf ("\nSum = %d", Sum);
    return 0;
}
```



Draw a Concept map for data structure and its classification

Data Structure

A data structure is a way of organizing and storing data in a computer so that it can be accessed and modified.

Classification

1) Primitive (Fundamental) Data Structures

Basic data types provided by the programming language

Examples

- * Integer
- * Float/Double
- * character
- * Boolean

2) Non-Primitive (Derived) Data Structures

Constructed from primitive data types. Further classified into:

Linear Data Structure

Elements are arranged in a sequential order (Each element except the first has one predecessor and except the last has one successor).

Examples

- * Array
- * Stack
- * Queue

Non-Linear Data Structure

Elements are not arranged in a sequential order. One element may be connected to multiple others.

Examples

- * Trees
- * Graphs